

Lab 1

The screenshot shows a database query tool interface. The 'Query' tab is active, displaying a SQL query:

```
1 SELECT rental_id, customer_id, return_date
2 FROM rental
3 WHERE return_date IS NULL
4 LIMIT 1;
```

Below the query, the 'Data Output' tab is active, showing the results of the query. The results are displayed in a table with columns: rental_id, customer_id, and return_date. The first row shows rental_id 11541, customer_id 335, and return_date [null].

	rental_id [PK] integer	customer_id smallint	return_date timestamp without time zone
1	11541	335	[null]

Chọn 1 rental để test

rental_id = 11541, customer_id = 335, return_date = null

The screenshot shows a database query tool interface. The 'Query' tab is active, displaying a SQL query:

```
6 BEGIN;
7 UPDATE rental
8 SET return_date = CURRENT_TIMESTAMP
9 WHERE rental_id = 11541;
```

Below the query, the 'Data Output' tab is active, showing the results of the query. The results are displayed in a table with columns: rental_id, customer_id, and return_date. The first row shows rental_id 11541, customer_id 335, and return_date 2026-09-18 15:05:04.645678.

	rental_id [PK] integer	customer_id smallint	return_date timestamp without time zone
1	11541	335	2026-09-18 15:05:04.645678

Update rental thành returned

The screenshot shows a database query tool interface. The 'Query' tab is active, displaying a SQL query:

```
11 SELECT rental_id, customer_id, return_date
12 FROM rental
13 WHERE rental_id = 11541;
```

Below the query, the 'Data Output' tab is active, showing the results of the query. The results are displayed in a table with columns: rental_id, customer_id, and return_date. The first row shows rental_id 11541, customer_id 335, and return_date 2026-09-18 15:05:04.645678.

	rental_id [PK] integer	customer_id smallint	return_date timestamp without time zone
1	11541	335	2026-09-18 15:05:04.645678

Chưa commit

```
15
16 INSERT INTO payment (customer_id, staff_id, rental_id, amount,
17 wrong_column)
18 VALUES (459, 1, 2, 5.00, CURRENT_TIMESTAMP);
```

Data Output Messages Notifications

ERROR: column "wrong_column" of relation "payment" does not exist
LINE 2: wrong_column)
^

SQL state: 42703
Character: 68

Intentionally "break" bằng cách sử dụng tên cột không tồn tại

```
19
20 ROLLBACK;
```

Data Output Messages Notifications

ROLLBACK

Query returned successfully in 56 msec.

```
11 SELECT rental_id, customer_id, return_date
12 FROM rental
13 WHERE rental_id = 11541;
14
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1

	rental_id [PK] integer	customer_id smallint	return_date timestamp without time zone
1	11541	335	[null]

Sau khi rollback => Huỷ luôn thay đổi ở Insert và Update

```

12 FROM rental
13 WHERE rental_id = 11541;
14
15
16 INSERT INTO payment (customer_id, staff_id, rental_id, amount,
17 payment_date)
18 VALUES (459, 1, 2, 5.00, CURRENT_TIMESTAMP);
19
20 SELECT payment_id, customer_id, staff_id, rental_id, amount,
21 payment_date
22 FROM payment
23 WHERE rental_id = 11541
24 ORDER BY payment_date DESC;
25

```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1 Page No: 1

	payment_id [PK] integer	customer_id smallint	staff_id smallint	rental_id integer	amount numeric (5,2)	payment_date timestamp without
1	31936	335	2	11541	0.99	2007-05-14 13:44:2

Nếu chạy đúng Insert và Update

Ex 1:

```

27 BEGIN;
28 -- Step 1 : Mark as returned
29 UPDATE rental SET return_date = NOW( ) WHERE rental_id = 2;
30 -- Step 2 : Intentional Error (Invalid payment
31 INSERT INTO payment (customer_id, staff_id, rental_id, amou
32 payment_date)
33 VALUES (459, 1, 2, 99.99, 'WRONG_DATE_FORMAT') ;
34 COMMIT;

```

Data Output Messages Notifications

WARNING: there is already a transaction in progress

ERROR: invalid input syntax for type timestamp: "WRONG_DATE_FORMAT"
 LINE 7: VALUES (459, 1, 2, 99.99, 'WRONG_DATE_FORMAT') ;
 ^

SQL state: 22007
 Character: 291

```
36 SELECT return_date FROM rental WHERE rental_id = 2 ;
```

Data Output Messages Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

Transaction aborted => return_date không được giữ lại

```
39 BEGIN;
40     INSERT INTO customer (first_name, last_name, email, address
41     VALUES ('AI', 'Student', 'ai@edu.vn', 1, 1);
42
43     SAVEPOINT user_created;
44
45     -- Try an experimental feature (may fail)
46     INSERT INTO special_offers (customer_id, code)
47     VALUES (9999, 'FREE');
48
49     -- If it fails:
50     ROLLBACK TO user_created;
51
52     -- The user is still in the "sandbox", ready to be committed!
53 COMMIT;
```

Data Output Messages Notifications

ERROR: relation "special_offers" does not exist
LINE 8: INSERT INTO special_offers (customer_id, code)
 ^

SQL state: 42P01

Character: 211

Tạo Savepoint => Để rollback về được

```
48
49     -- If it fails:
50     ROLLBACK TO user_created;
51
52     -- The user is still in the "sandbox", ready to be comm
53 COMMIT;
```

Data Output Messages Notifications

ROLLBACK

Query returned successfully in 49 msec.

Ex 2:

```
55
56 BEGIN;
57 -- 1. Create a new staff member
58 INSERT INTO staff (first_name, last_name, address_id, email,
59     store_id, username)
60 VALUES ('AI', 'Student', 1, 'ai.staff@edu.vn', 1, 'ai_student'
61 );
62
63 -- 2. Create a savepoint AFTER staff has been created
64 SAVEPOINT staff_created;
65
66 -- 3. Intentionally assign the staff to a non-existent store
67 UPDATE staff
68 SET store_id = 9999
69 WHERE username = 'ai_student';
```

Data Output Messages Notifications

WARNING: there is already a transaction in progress
SAVEPOINT

Query returned successfully in 51 msec.

```
66 -- 3. Intentionally assign the staff to a non-existent store
67 UPDATE staff
68 SET store_id = 999999
69 WHERE username = 'ai_student';
70
71 -- This should fail because store_id = 9999 does not exist
72
73 -- 4. Roll back ONLY to the savepoint
74 ROLLBACK TO SAVEPOINT staff_created;
75
```

Data Output Messages Notifications

ERROR: smallint out of range

SQL state: 22003

Fail

```
73 -- 4. Roll back ONLY to the savepoint
74 ROLLBACK TO SAVEPOINT staff_created;
75
```

Data Output Messages Notifications

ROLLBACK

Query returned successfully in 70 msec.

Rollback về Savepoint

```
76 -- Staff still exists, now assign the correct store
77 UPDATE staff
78 SET store_id = 1
79 WHERE username = 'ai_student';
80
81 -- 5. Save everything
82 COMMIT;
83
84
85
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 35 msec.